THE DIAGONAL TENSOR-INTERNAL-HOM MAP IS AN ISOMORPHISM FOR A PERFECT ARGUMENT OR FOR A PSEUDO-COHERENT ARGUMENT WITH BOUNDED BELOW SECOND OBJECT AND TOR AMPLITUDE CONDITION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The eight references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- K-injective complexes and the derived tensor product as a total complex. used throughout the counted proof span. Supplies the explicit complexes between which the map is compared.
- Finitely generated projective modules and the tensor-evaluation adjunction. used in the counted proof span. Is the reason each component of the displayed map is an isomorphism.
- Boundedness and finite injective dimension as hypotheses controlling the indices. the two cases in the statement and their use in the counted proof span. Is what makes the finiteness bookkeeping work in each case.

Source reference. Original definition 15874: sources/stacks-more-algebra/pinned-original-source.tex, source lines 15874–15874. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 0ATH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13615–13615. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 18708: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18708–18708. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 16930: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16930–16930. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 18036: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18036–18036. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 18799: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18799–18799. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 20288: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20288–20288. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let K, L, M be objects of $D(R)$. The map*

$$K \otimes^{\mathbf{L}}_R R\mathrm{Hom}_R(M, L) \longrightarrow R\mathrm{Hom}_R(M, K \otimes^{\mathbf{L}}_R L)$$

of Lemma 18799 is an isomorphism in the following cases

1. *M perfect, or*
2. *K is perfect, or*
3. *M is pseudo-coherent, $L \in D^+(R)$, and K has tor amplitude in $[a, \infty]$.*

Proof. Proof in case M is perfect. Note that both sides of the arrow transform distinguished triangles in M into distinguished triangles and commute with direct sums. Hence it suffices to check it holds when $M = R[n]$, see Derived Categories, Remark 0ATH and Lemma 20288. In this case the result is obvious.

Proof in case K is perfect. Same argument as in the previous case.

Proof in case (3). We may represent K and L by bounded below complexes of R-modules $K^\bullet$ and $L^\bullet$. We may assume that $K^\bullet$ is a K-flat complex consisting of flat R-modules, see Lemma 16930. We may represent M by a bounded above complex $M^\bullet$ of finite free R-modules, see Definition 15874. Then the object on the LHS is represented by

$$\mathrm{Tot}(K^\bullet \otimes_R \mathrm{Hom}^\bullet(M^\bullet, L^\bullet))$$

and the object on the RHS by

$$\mathrm{Hom}^\bullet(M^\bullet, \mathrm{Tot}(K^\bullet \otimes_R L^\bullet))$$

This uses Lemma 18708. Both complexes have in degree n the module

$$\bigoplus\nolimits_{p+q+r=n} K^p \otimes \mathrm{Hom}_R(M^{-r}, L^q) = \bigoplus\nolimits_{p+q+r=n} \mathrm{Hom}_R(M^{-r}, K^p \otimes_R L^q)$$

because M^{-r} is finite free (as well these are finite direct sums). The map defined in Lemma 18799 comes from the map of complexes defined in Lemma 18036 which uses the canonical isomorphisms between these modules. □